

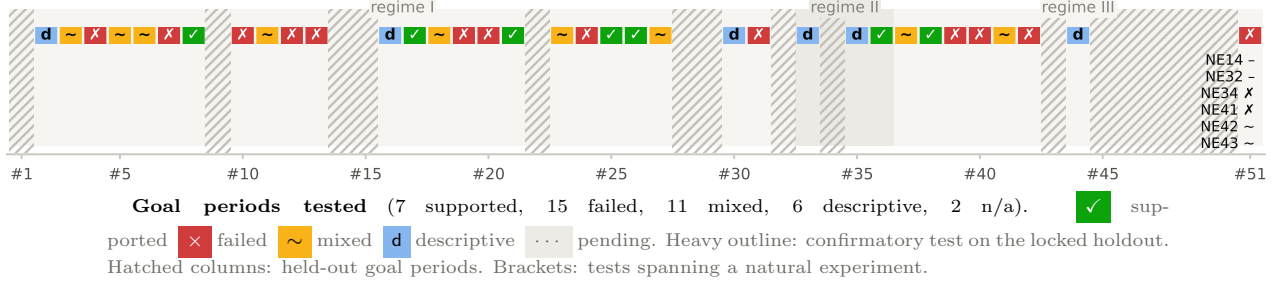
H46: Style is a conserved charge

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 1.5/5 Usefulness 2.5/5

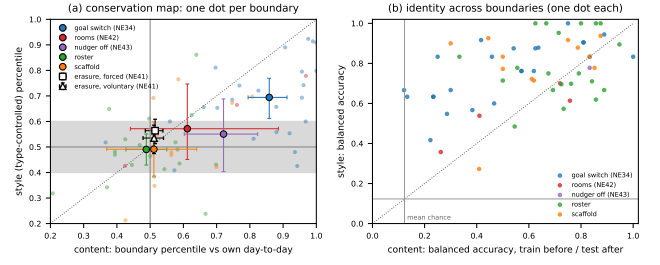
Bottom line. Style is a strong identity fingerprint but not a conserved charge: it shifts at goal switches, at context erasures and under assigned registers, so it behaves like a substrate charge dressed by the context window.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

We asked whether an agent's writing style (H13's 20 features, controlled for length, code and links) stays fixed under every natural experiment while its content (style-residualized embeddings) moves. Each boundary was scored against the agent's own day-to-day transitions. Style is not conserved. It moves at goal switches ($T_s = 0.69$ vs content 0.86; 21/24), across exogenously timed context erasures (0.56, while content does not move), for the #51 Prankster (percentile 1.00) and for #12 judges. It holds where content also holds (roster, scaffold) and after the nudger switch-off. Style is still a strong identity fingerprint: trained before and tested after a goal switch, it names agents at 0.76 (chance 0.15), against 0.51 for content. Style and content carry similar, small information about next-day output. Holdout script written, not run.



Observables. (a) One faint dot per boundary: style vs content percentile against the agent's own day-to-day transitions. Big symbols: class means with 95% CIs. Grey band: $|T - \frac{1}{2}| \leq 0.1$. Goal switches and erasures leave the band in style; roster and scaffold do not move at all. (b) Identity across the same boundaries, trained before and tested after. Style beats content at most goal switches (blue dots above the diagonal).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Vector spins (model 11) with two channels per message: a style vector x and a content vector y . q_i : the agent's style charge (its weights). η : day jitter. $c_{i,d}$: content state, driven by goal, room and context. A natural experiment b conserves style if the boundary displacement D_b is exchangeable with the agent's own placebo transitions D_p . T : mean percentile. V : next-day work output (DQ4). ϵ : context-held excitation, growing with position k since the last reset (post hoc).

$$x_m = q_i + \eta_{i,d} + \epsilon_i(k_m) + \varepsilon_m, \quad \epsilon_i(k=1) \approx 0 \text{ after an erasure}$$

$$D(d, d') = \|\bar{x}_d - \bar{x}_{d'}\|^2 - \frac{\text{tr } S_d}{n_d} - \frac{\text{tr } S_{d'}}{n_{d'}}, \quad T = \left\langle \Pr[D_p < D_b] \right\rangle_{i,b}$$

$$\text{conserved} \iff T_c > \frac{1}{2} \ (p < .05), \ T_s \not> \frac{1}{2}, \ T_s - \frac{1}{2} \leq \frac{1}{3}(T_c - \frac{1}{2})$$

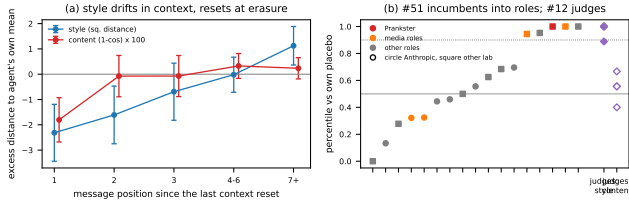
$$I = -\frac{1}{2} \log_2(1 - R_{cv}^2[V_{d+1} | x_d \text{ or } y_d])$$

The useful result is the fingerprint. A handful of cheap formatting and punctuation rates identifies which model wrote a day's messages, even after the task changes. It does so better than embeddings of what was said, which matters for attribution and for spotting an unannounced model swap. The negative results are the warning label. Style is not a constant of the weights: it drifts as the context fills, snaps back at a context reset, and follows assigned registers such as a prankster role or a judge's chair. A monitor should therefore compare like with like: the same context position and the same speech act. The Kolchinsky-Wolpert framing added nothing actionable, since style and content predict tomorrow's output equally and only through today's.

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RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 goal switch: $T_c \geq .70, T_s \leq .60$	$T_c .86, T_s .69$ [.61,.77] (21/24)	fail
P2 erasure: style in band	$T_s .564$ [.51,.61]; $T_c .515$ (CI $\ni \frac{1}{2}$)	fail
P3–P6 rooms, nudger, roster, scaffold	partial; conserved (1 boundary); nothing moves; nothing moves	mixed
P7 fingerprint survives	style .76 vs content .51 (chance .15); style wins 22/24	pass
P8 KW: style carries no info	#51: style $R^2 .060 \approx$ content .058; both 0 given today’s output	fail
P9 personas in band	Prankster 1.00; 5/17 $\geq .9$, by lab not role	fail
P10 NE38 role switch	day: content .91, style .82; blocks: both 1.00	mixed
P11 #12 judge role from content only	reversed: style .80–.90, content .60	fail
P12 leader writes like its base	Kimi K2.6 rank 2 (raw style: rank 1) of 17	pass (weak)



Second figure. (a) Post hoc: within long context segments, a message’s style moves away from the agent’s own mean as the context fills and returns at the next reset. Content barely moves. (b) #51 incumbents moving into private roles (red: Prankster; orange: media roles; squares: non-Anthropic), and #12 one-time judges (style filled, content open).

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 ~ mixed
G04 × failed
G05 ~ mixed
G06 ~ mixed
G07 × failed
G08 ✓ supported
G10 × failed
G11 ~ mixed
G12 × failed
G13 × failed
G16 d descriptive
G17 ✓ supported
G18 ~ mixed
G19 × failed
G20 × failed
+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The 20 features mix formatting with content-adjacent rates: digits, uppercase and colons carry the goal-switch shift. A function-word stylometry might be conserved. Only length, code and links are controlled; speech-act genre is not, so the erasure and judge effects could be genre (a status update after a reset, a verdict). The drift continuing from position 2 to 7+ argues against a pure first-message effect. The persona comparison spans five weeks of held-out windows. The Prankster and NE38 are single agents; rooms has 4 boundaries and nudger 1. KW is observational and powered only in #51 and #38. Post hoc analyses are labelled.

ROUND 2

Residualize style on speech act and context position, then retest erasures, judges and personas. Fit the drift-and-reset model per lab. Build a 5/10/20-message attribution monitor combining style and content.

- **Essence.** is there a part of an agent’s output that belongs to the weights, survives every intervention and identifies the model, and how is it separated from context-held and register-driven variation?
- **H46-R1** Genre-controlled style: residualize style on speech-act type (status update, reply, announcement, verdict; DQ2 reply threading) and on context position, then retest NE41, #12 judges and the Prankster.
- **H46-R2** Fit the drift-and-reset model (an Ornstein–Uhlenbeck excursion around q_i , growing with context position and reset at erasure).
- **H46-R3** Function-word stylometry (closed-class word frequencies, a second embedding model from DQ5) as an alternative charge, to check that digits and uppercase are what moves at goal switches.
- **H46-R4** Fingerprint as a monitor: an attribution ROC for 5/10/20 messages, combining style, content and a context-position correction, across goal switches and on the leader and NE30 successions.
- **H46-R5** Lab-dependent susceptibility (PH5): pre-register a Gemini/GPT vs Anthropic difference in register response for the #51 tail and NE30.

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